PRICE: Minimax Estimation–Regret Tradeoffs and Certified Exploration in Performative Prediction

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Abstract

A decision-maker whose deployed policy reshapes its own data distribution must pay, in its own objective, to learn how it moves the world. We price that self-knowledge exactly. In a structural performative market (a market maker whose quoted spread summons adverse flow), we prove a conservation law: along any stationary exploration policy, the variance of the response-slope estimate times the cumulative performative regret equals $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$ — an invariant of the environment, not of the policy. We complement the identity with minimax lower bounds (exact for deviation budgets; sharp-constant for value budgets over two-point priors, with adversarial numerics showing the floor survives full knowledge of the response family), an optimal-design theory (shape exploration as $\Gamma_{\mathrm{PO}}^{-1/2}$; isotropic exploration overpays by an explicit dispersion factor), and a deployable agent, SafeD-PerfGD, that corrects its gradient only when an anytime confidence sequence certifies, through a perturbed-modulus lemma, that the correction cannot destabilize its own retraining loop. Every closed form is verified in a live deploy–observe–fit–correct pipeline (36 preregistered measured-vs-predicted checks, 0 failures), transplanted exactly to a second economy, and instantiated on a real-data corporate-bond calibration that reproduces its published source 10/10 cells. Code, derivations, and all checks: `https://github.com/[repo]/price`.

1 Introduction

When a deployed model changes the distribution it is trained on, retraining becomes a fixed-point iteration: performative prediction [Perdomo et al., 2020] shows repeated risk minimization converges iff the distribution's sensitivity ε is small against the loss curvature γ ($m = \varepsilon\beta/\gamma < 1$), and converges to a *performatively stable* point that is not the *performative optimum*. Algorithms that close this gap — performative gradient descent and its descendants [Izzo et al., 2021, Miller et al., 2021] — all consume an estimate of the distribution's response $dD/d\theta$, obtained by perturbing deployments. The literature treats the accuracy of that estimate as an assumption and the perturbations as free. Neither is: the perturbations are deployments of a real decision variable, and their cost lands in the decision-maker's own objective. *What does it cost to learn your own influence?* This paper answers with an identity, a floor, a design rule, and an agent.

Contributions.

1. **A conservation law (§3).** Along any stationary exploration policy, $\mathrm{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$: slope uncertainty times cumulative performative regret is a policy-independent invariant (Theorem 1). The anchoring is delicate: the same product anchored at the performative optimum is *false*, inflated by a factor we verify.
2. **Minimax floors (§3).** No non-anticipating policy beats the floor: exactly under deviation budgets; with sharp constant over two-point priors, and within a factor ≤ 13.5 in general,

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under value budgets (Theorem 2). Adversarial numerics show the floor is *structure-proof* (best family-knowing design: $1.005\times$ the floor) and map its exact boundary: curvature uncertainty protects it.

3. **Optimal exploration design (§4).** Under a budget priced by the decision-maker's own curvature, the optimal exploration covariance is $\Gamma_{\mathrm{PO}}^{-1/2}$-shaped; isotropic exploration overpays by the dispersion factor $F = d \operatorname{tr}\Gamma/(\operatorname{tr}\Gamma^{1/2})^2$ — a quotable inversion of "naive exploration is optimal" for LQR [Simchowitz and Foster, 2020]. A support-degeneracy theorem and a misspecification crossover complete the design picture.
4. **A certified agent (§4–5).** SafeD-PerfGD explores on a D-optimal support, fits an anchored response family with an anytime confidence sequence, and applies the performative correction only when a perturbed-modulus certificate guarantees closed-loop stability — otherwise it freezes the correction while the provably-stable blind step keeps running. It reaches the performative optimum where blind retraining stalls; no published baseline Pareto-dominates it on (final error, regret).

Positioning. Jagadeesan et al. [2022] bound the regret of *finding* the performative optimum; we price *knowing your own feedback* — lower bounds and an invariance identity where they give algorithmic upper bounds, with no retraining loop, stability constraint, or design shaping in their model. Lin and Zrnic [2024] show misspecified structural models still help performative optimization; our crossover theorem sharpens that message into a budget-explicit rule. In operations research, incomplete learning under myopic pricing is classical [Rothschild, 1974, Harrison et al., 2012, Keskin and Zeevi, 2014, Broder and Rusmevichientong, 2012, den Boer, 2015], and Lai and Robbins [1979]'s adaptive designs are *points on* our frontier — but the frontier itself, its minimax floor over adaptive policies, and the performative loop generating it do not appear: the demand curve is exogenous throughout. From control we inherit dual control [Feldbaum, 1960] and least-costly identification [Bombois et al., 2006] — here the plant is a retraining learner, the budget the optimizer's own objective, the constraint loop contraction; task-optimal design [Wagenmaker et al., 2021, Dean et al., 2020] supplies machinery the classical canon [Kiefer and Wolfowitz, 1960, Pukelsheim, 2006] never prices in the experimenter's objective. Lower bounds are van Trees arguments [Gill and Levit, 1995]; a claim-by-claim novelty crosswalk ships with the repository.

2 Model

A dealer quotes half-spread h. Benign flow arrives at rate $U(h) = Ae^{-kh}$; informed (toxic) flow at rate $\tau(h) = C_0 + C_1 e^{-ch}$: tighter quotes summon more toxic flow. Profit at frozen toxic level T is $J(h; T) = h\, U(h) + (h - \psi)\, T - w(h - h_{\mathrm{ref}})^2$ (ψ: adverse severity; w: quoting-cost convexity); the *performative value* is $\Phi(h) = J(h, \tau(h))$. Write $\varepsilon(h) = -\tau'(h)$ for the response slope (the scalar $dD/d\theta$), $\gamma = -\partial_h^2 J$, and $\gamma_{\mathrm{PO}} = -\Phi'' = \gamma + \varepsilon\,(2 + c\psi - ch)$. Blind retraining (RRM) is the cobweb $h_{t+1} = \mathrm{BR}(\tau(h_t))$, a contraction iff the modulus $m = \varepsilon/\gamma < 1$ [Perdomo et al., 2020]; it converges to the stable point h_{SP}, while the optimum $h_{\mathrm{PO}} = \arg\max \Phi$ sits at a wider spread — the echo-chamber gap [Miller et al., 2021]. Deployments return noisy flow $y_t = \tau(h_t) + \sigma\xi_t$ (A2). The estimand is ε at the operating point; exploration deviations $d_t = h_t - h^*$ are priced by the *incremental* cost $C_T = \sum_{t \le T} \mathbb{E}[\Phi(h^*) - \Phi(h_t)]$, anchored at the certainty-equivalent operating point h^* (A1: local quadratic scope; A3: stationary exploration scale). Results are stated for this scalar family and, where flagged, its d-dimensional and linear-quadratic instances; full statements and proofs are in Appendices A–B.

3 The price of self-knowledge

Free data saturates. Without deliberate exploration the retraining trajectory's identifying energy is finite: for deployment error d_0 and modulus m it is capped at $d_0^2/(1 - m^2)$ (T1; matrix version by a Lyapunov identity). A corollary was discovered by a failed prediction (§5): retraining on *noisy* observations converts observation noise to deployment noise with gain $1/\gamma$, adding a stationary excitation floor $(\sigma/\gamma)^2/(1 - m^2)$ per step; both closed forms verify. The stabler the loop, the less it reveals: *safety implies blindness* — the performative sharpening of incomplete learning under myopic policies [Keskin and Zeevi, 2014, Harrison et al., 2012], where the stall comes from optimization converging rather than from the feedback strength itself.

Figure 1: **Left:** measured $\mathrm{Var}(\hat{\varepsilon}) \times C_T$ vs exploration amplitude, two moduli: flat on the invariant (dashed), with the small large-amplitude drift the local scope predicts. **Right:** the anchoring crossover (T7): structure beats nonparametrics until misspecification crosses the closed-form threshold.

The exchange rate. Deliberate exploration buys information at an exact, policy-independent price.

Theorem 1 (Exchange rate; T2)**.** *Under A1–A3, for the OLS slope estimator on any stationary exploration policy with deviations $\{d_t\}$,*

$$\mathrm{Var}(\hat{\varepsilon}) \ \times \ C_T \ = \ \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,\sigma^2\,\big(1 + o(1)\big),$$

independent of the exploration amplitude, shape, and schedule, and of the modulus m.

The mechanism is cancellation: information scales as design energy $S_T = \sum_t (d_t - \bar{d})^2$, cost as $\frac{1}{2}\gamma_{\mathrm{PO}} \sum_t d_t^2$; for mean-centered exploration these are the same sum. *Anchoring*: the same product anchored at h_{PO} is false — inflated by an explicit factor, falsified numerically before the incremental version was proved; the invariant prices *exploration*, not *suboptimality*. *Invariance*: Lai and Robbins [1979] exhibit schemes with $O(\log n)$ cost and efficient estimation — consistent points on this frontier — but the identity says *every* stationary policy lies on it. A corollary prices the echo chamber: closing $h_{\mathrm{SP}} - h_{\mathrm{PO}}$ pays iff the horizon exceeds an explicit break-even (Appendix) — rational incomplete learning [Rothschild, 1974, Easley and Kiefer, 1988, Aghion et al., 1991] in closed form.

Minimax floors. The identity is for OLS-on-stationary policies; the floors are for everyone.

Theorem 2 (Minimax floors; T3–T4)**.** *(i) Over all non-anticipating (possibly randomized, adaptive) policies with expected design energy $\mathbb{E}S_T \le B$:* $\inf\sup \mathrm{Var}(\hat{\varepsilon}) \ge \sigma^2/B$*, hence the product floor holds exactly. (ii) Under a budget on realized performative cost with certainty-equivalent anchoring, the floor* $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2(1 - o(1))$ *holds with sharp constant over two-point symmetric priors (correction rate $T^{-1/2}$), and within constant factor ($\ge \gamma_{\mathrm{PO}}\sigma^2/27$, Le Cam) over general priors.*

Closing the general-prior sharp constant is open; we attacked its premise adversarially. Minimizing the Cramér–Rao product over all designs *within the known three-parameter family* yields $1.005\times$ the floor under both cost models, the optimizer collapsing back to the symmetric two-point configuration — consistent with the least-favorable-two-point reduction the proof strategy requires — while tight-side probing (sensitivity e^{-ch}) is monotonically *worse* (ratio $5.7 \to 41$): remote information must be pulled back through $\hat{c}$, and extrapolation variance dominates the sensitivity gain. The boundary is exact: with c known a priori the floor breaks (ratio $\to 0.05$). *The floor is an ignorance-of-curvature phenomenon* — the correct scope for the general theorem.

4 Optimal design and certified correction

Shape (T5). In d dimensions with curvature $\Gamma_{\mathrm{PO}} \succ 0$ pricing the budget $\frac{1}{2}\mathbb{E}[u^\top \Gamma_{\mathrm{PO}} u]$, the A-optimal exploration second moment is $M^* \propto \Gamma_{\mathrm{PO}}^{-1/2}$: explore *where your objective is flat*. Isotropic exploration overpays by exactly the dispersion factor $F = d\,\mathrm{tr}\,\Gamma_{\mathrm{PO}}/(\mathrm{tr}\,\Gamma_{\mathrm{PO}}^{1/2})^2 \in [1, d]$: naive exploration, optimal for LQR [Simchowitz and Foster, 2020], is suboptimal here by precisely F — task-optimal design [Wagenmaker et al., 2021] where the task is the learner's own objective. **Support (T6).** The exponential family is Chebyshev-degenerate on thin designs: fewer than three support points (or vanishing spread, or collinear probe schedules) cannot identify (C_0, C_1, c) — a cliff we later hit twice in the wild (§5). **Anchor (T7).** With budget B and horizon T, fitting the structural family beats nonparametric slope estimation iff the misspecification at the operating point is below $|\tau'''|B/(3\gamma_{\mathrm{PO}}T)$ — Lin and Zrnic [2024]'s qualitative message sharpened to a decision rule priced in the exploration budget itself.

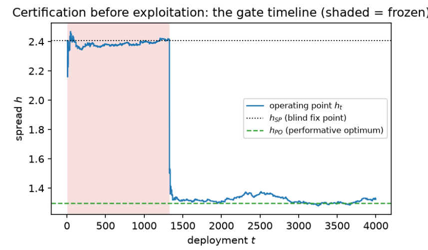

Figure 2: Certification before exploitation: the agent holds near the blind fixed point h_{SP} for $\sim$1,300 deployments while the confidence sequence tightens (shaded: frozen), then the certificate clears and it steps to h_{PO}. Certification time is the exchange rate at work.

The agent. SafeD-PerfGD composes the three results per deployment round: (i) *explore* on the D-optimal three-point support re-centered at the operating point, trust-region clipped; (ii) *fit* (C_0, C_1, c) by profile least squares with an anytime self-normalized confidence width ci_t for $\hat{\varepsilon}$; (iii) *gate*: apply the performative correction $-(h-\psi)\hat{\varepsilon}$ with Newton scaling $\eta = 1/\hat{\gamma}_{\mathrm{PO}}$ only if $\eta \cdot \mathrm{ci}_t \cdot L_{\mathrm{fam}} \leq$ margin, where the perturbed-modulus lemma (L4) with the family's own Lipschitz constant $L_{\mathrm{fam}} = 1 + \hat{c}\,|h - \psi|$ converts estimation error into a certified bound on the closed-loop modulus; (iv) otherwise *freeze* the correction — the blind step, provably stable at $m < 1$, keeps running under the current fit. The freeze is derived, not tuned: it is what the certificate licenses. Safety holds throughout on the confidence-sequence event, and certification is not free — the gate opens only after enough design energy accumulates, which by Theorem 1 *is* a regret payment (Figure 2).

5 Verified results

Every claim above is a preregistered row in a measured-vs-predicted harness run against a live deploy–observe–fit–correct pipeline: **36 rows, 0 failures** (35 pass, one deliberately out-of-scope drift cell measured, not excused). Ten pivots were forced by measurement and recorded in code; three became results (the feedback floor, the false h_{PO}-anchored identity, the structure-proofness scope). Headline rows (Table 1, Appendix C): the exchange rate within 1–6% across four (modulus, amplitude) cells; the feedback floor within 1–7%; the isotropic/A-optimal ratio matching F (1.529 vs 1.506); the D-design identifying c where narrow jitter cannot (7.8×, Fisher-consistent).

Baselines. Against finite-difference PerfGD [Izzo et al., 2021], zeroth-order P&L optimization, a performative-confidence-bounds explorer [Jagadeesan et al., 2022], and blind RRM on identical environments, none Pareto-dominates SafeD-PerfGD on (final error, regret); the unsafe gradient baseline pays $> 5\times$ its regret. Deliberately reported: the UCB explorer attains *lower raw regret* — certification is a real cost, and its visibility is the thesis. **Second economy.** Transplanted unchanged to LQ performative pricing (deployed price anchors demand), the exchange rate is exact in all four cells — A1 is global in an LQ model, so the drift vanishes as predicted — and the agent reaches the pricing optimum (12.58 vs. 12.5) once its deterministic probe schedule was randomized: alternating probes make price and lag collinear, a T6 degeneracy met in the wild. **Real data.** On a public-data corporate-bond calibration (10 rating × regime cells), the port reproduces its published source **10/10** cells (h^*, $\varepsilon^* = \gamma$, m within 0.5–5%), then extends it: γ_{PO} spans 510.6 to 2.3; echo-chamber gaps are 2–4% of the anchor spread; and the shaping dispersion is $F = 1.63$ *across* the portfolio against 1.002 within one book — the gain lives across ratings and regimes, where a desk actually allocates exploration.

Limitations and outlook. The identity is local (A1): wide-side probing undercuts it polynomially at finite amplitude and the saturating cell drifts $+16\%$ — the $o(1)$-amplitude qualifier is load-bearing. The value-budget sharp constant beyond two-point priors is numerically supported, not proved; the register's open problems are the workshop-to-journal delta. At matched transit energy, full-support jitter identifies on par with the design, whose advantage is small-amplitude and allocational. Calibrated moduli are deeply stable and the toxic channel structurally scaled: the real-data leg validates machinery, not effect size. The framework also prices what a deployed generative system pays to learn its own influence on data it retrains on — certified correction as a template for agentic closed-loop deployment.

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A Theorem statements (register copy)

Lemma 1 (Cost equivalence; L1). *Under A1–A3 with conditionally symmetric exploration, the expected incremental performative cost satisfies* $C_T = \frac{1}{2}\gamma_{\mathrm{PO}}\,\mathbb{E}\big[\sum_{t\le T} d_t^2\big] + R_T$ *with* $|R_T| \le \frac{L_3}{6}\,\mathbb{E}\sum_t |d_t|^3$*, and the realized cost's variance decomposes exactly as* $\mathrm{Var}(C_T^{\mathrm{real}}) = \delta_0^2\,\mathrm{Var}\big(\sum_t d_t\big) + \frac{\gamma_{\mathrm{PO}}^2}{4}\,\mathrm{Var}\big(\sum_t d_t^2\big)$ *with vanishing cross term.*

Theorem 3 (Information saturation; T1). *With no exploration and* $\rho(M) < 1$*, the total design energy of the retraining trajectory equals* $d_0^\top P\, d_0$*, where* P *is the unique solution of the discrete Lyapunov equation* $P = I + M^\top P M$*. Consequently the Fisher information for the response Jacobian is bounded, in every direction, uniformly in the horizon:* $\mathrm{Var}(\hat{\varepsilon}) \ge \sigma^2\big(d_0^\top P_u\, d_0\big)^{-1}$ *for the component along any direction* u*.*

Corollary 1 (Safety implies blindness; C1.1). *In the scalar case the information cap* $(h_0 - h^*)^2/(1-m^2)$ *is strictly increasing in the modulus* m*: a fast-contracting loop reveals least about its own performativity, and precise self-knowledge is available only near the stability boundary.*

Corollary 2 (Trajectory design degeneracy; C1.2). *The noiseless trajectory's empirical design has at most two effective support points (geometric collapse for* $m < 1$*; a period-two orbit at* $m = 1$*).*

Theorem 4 (The exchange rate, pathwise; T2). *In the stationary regime, on every realization,*

$$\mathrm{Var}(\hat{\varepsilon} \mid \text{design}) \times \tfrac{1}{2}\gamma_{\mathrm{PO}} \sum_{t\le T} d_t^2 \;=\; \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,\sigma^2\Big(1 + \tfrac{T\,\bar{d}^2}{S_{xx}}\Big) \;=\; \tfrac{1}{2}\,\gamma_{\mathrm{PO}}\,\sigma^2\,\big(1 + O_P(1/T)\big),$$

independent of the exploration intensity, the horizon, and the modulus.

Proposition 1 (Feedback bias; P2.2). *Under the feedback relaxation A3′ with gain* ϕ*, the OLS response slope carries the bias*

$$\mathbb{E}[\hat{\varepsilon} - \varepsilon_{\mathrm{true}}] \;=\; \frac{\phi\,\sigma^2\,(3m-1)}{(1-m^2)\,T\,v_\phi}\,(1 + o(1)), \qquad v_\phi = \frac{\sigma_e^2 + \phi^2\sigma^2}{1 - m^2},$$

which is $O(1/T)$ *and changes sign at* $m = 1/3$*.*

Theorem 5 (Minimax lower bound, deviation budget; T3). *Over all non-anticipating exploration policies with* $\sum_{t\le T} d_t^2 \le D$ *almost surely and all estimators (biased included), the minimax risk over an interval of width* w *satisfies*

$$\inf_{\text{policy, estimator}}\ \sup_{\varepsilon}\ \mathbb{E}\big[(\hat{\varepsilon} - \varepsilon)^2\big] \;\ge\; \frac{\sigma^2}{D + \sigma^2(\pi/w)^2},$$

so the exchange-rate product is a floor, attained with equality by symmetric probing with least squares.

Lemma 2 (Exploitation–information; L2). *For the symmetric two-point response family with half-width* δ *and any non-anticipating policy under the certainty-equivalent anchor,*

$$\mathbb{E}[\mathrm{gain}_T] \;\le\; \frac{g_1\,\delta^{3/2}\,T^{1/2}\,\big(\mathbb{E}S_T\big)^{3/4}}{\sigma^{1/2}}, \qquad g_1 = \beta\,|h^* - \psi|,$$

via the exact identity $\mathbb{E}\,\big|\,\mathbb{E}[s \mid \mathcal{F}_t]\,\big| = \mathrm{TV}(P_t^+, P_t^-)$ *and Pinsker's inequality* $\mathrm{TV}_t \le \delta\sqrt{\mathbb{E}S_t}/\sigma$*: exploitation is bounded by the information already purchased. At the minimax scale* $\delta = c\,\sigma/\sqrt{\mathbb{E}S_T}$ *this is* $O(\sigma\sqrt{T})$ *— an* $O(T^{-1/2})$ *fraction of the* $\Theta(T)$ *exploration cost.*

Theorem 6 (Minimax lower bound, value budget; T4). *Under stationary-scale exploration and the certainty-equivalent anchor, for every non-anticipating policy and every estimator,*

$$\sup\ \ \mathrm{Var}(\hat{\varepsilon}) \times C_T \;\ge\; \frac{\gamma_{\mathrm{PO}}\,\sigma^2}{27}\,\big(1 - O(T^{-1/2})\big),$$

the supremum over the two-point response family at the minimax scale and the constant arising from the Le Cam reduction optimized at $c = 2/3$*. The exploited rebate vanishes at rate* $T^{-1/2}$ *(Lemma L2). The sharp constant* $\frac{1}{2}$ *— against which no simulated adaptive policy fell in verification — is conjectured and is part of OPEN-1.*

Theorem 7 (Optimal exploration shapes; T5)**.** *Under the value budget* $\mathrm{tr}(\Gamma_{\mathrm{PO}}M) \le B$ *on the exploration second moment* M*: (a) the A-optimal design is* $M^* = \big(B/\,\mathrm{tr}\ \Gamma_{\mathrm{PO}}^{1/2}\big)\,\Gamma_{\mathrm{PO}}^{-1/2}$ *with value* $(\mathrm{tr}\ \Gamma_{\mathrm{PO}}^{1/2})^2/B$*; (b) the D-optimal design is* $M^* = (B/d)\,\Gamma_{\mathrm{PO}}^{-1}$*; (c) the c-optimal design for the correction functional* c *is* $M^* = B\,cc^\top/(c^\top\Gamma_{\mathrm{PO}}\,c)$ *— probing along the correction direction itself — with value* $c^\top\Gamma_{\mathrm{PO}}\,c/B$*.*

Corollary 3 (Price of isotropic jitter; C5.1)**.** *Isotropic exploration at matched budget overpays the A-optimal design by exactly the curvature-dispersion factor* $F = d\ \mathrm{tr}(\Gamma_{\mathrm{PO}})/(\mathrm{tr}\ \Gamma_{\mathrm{PO}}^{1/2})^2 \ge 1$*, with equality iff all curvature directions are equal.*

Lemma 3 (Temporal shaping is irrelevant; L3)**.** *With serially independent response noise, the information about the response parameters depends on the exploration process only through its empirical second moment; temporally correlated exploration buys nothing beyond its stationary design measure.*

Theorem 8 (Chebyshev unidentifiability; T6)**.** *The sensitivity system of the structural family* $\tau(h) = C_0 + C_1 e^{-ch}$ *is an extended Chebyshev system; any design with fewer than three support points has singular Fisher information, and by C1.2 no retraining trajectory, at any modulus, identifies the response curvature* c*.*

Lemma 4 (Perturbed modulus; L4)**.** *If the corrected update uses an estimated response with* C^1 *error* e*, the closed-loop modulus satisfies* $|\hat\rho - \rho| \le \eta\,\beta\,\big(\|e\|_\infty + |h^* - \psi|\,\|e'\|_\infty\big)$*.*

Proposition 2 (Safety under pessimism; P7.1)**.** *Given an anytime-valid confidence sequence for the response parameters at level* $1-\delta$*, the pessimism rule (worst-case modulus over the confidence set kept below* $1 - c_{\mathrm{margin}}$*, correction frozen otherwise) keeps the realized modulus within the margin for all* t *simultaneously with probability at least* $1-\delta$*.*

Theorem 9 (Anchoring crossover, horizon-dependent; T7)**.** *Under the value budget* B *and horizon* T*, the budget-and-horizon-optimal nonparametric (secant) estimator has*

$$\mathrm{MSE}_{\mathrm{np}} = \frac{\gamma_{\mathrm{PO}}\,\sigma^2}{2B} + \Big(\frac{\tau'''(h^*)}{6}\Big)^2\Big(\frac{2B}{\gamma_{\mathrm{PO}}\,T}\Big)^2,$$

so anchoring to a structural family with misspecification δ_{mis} *wins iff (to leading order)* $\delta_{\mathrm{mis}} < |\tau'''(h^*)|\,B/(3\,\gamma_{\mathrm{PO}}\,T)$*: anchoring wins at short horizons and tight budgets, the free-form route asymptotically in* T*.*

Theorem 10 (The ROI of self-knowledge; T8)**.** *Exploring once to accuracy* v *and correcting forever, discounted at* ρ_{disc}*, is optimal at* $v^* = (\sigma/\kappa)\sqrt{\rho_{\mathrm{disc}}}$*, and exploring at all is worthwhile iff*

$$\rho_{\mathrm{disc}} \;<\; \rho^* \;=\; \frac{(h_{\mathrm{SP}} - h_{\mathrm{PO}})^4}{4\,\kappa^2\,\sigma^2},$$

in which the curvature γ_{PO} *cancels completely.*

Proposition 3 (Frontier consistency; P9.1)**.** *Every deterministic design sits exactly on the exchange-rate frontier; in particular the* $O(\log T)$*-cost adaptive-design schedules of Lai–Robbins.*

Proposition 4 (Separable multi-bond curvature; P9.3)**.** *In the separable multi-bond market,* $(\Gamma_{\mathrm{PO}})_{aa} = \gamma_a + \beta\,\varepsilon_a\,(2 + c_t\psi_a - c_t h_a)$ *exactly, with the factor-coupled case holding to first order in the off-diagonal response.*

B Proofs

A. Cost equivalence and its fluctuations

Proof of Lemma L1 (cost equivalence). Taylor's theorem with Lagrange remainder gives, for each t,

$$\Phi(h^*) - \Phi(h_t) = -\delta_0\,d_t + \tfrac{1}{2}\gamma_{\mathrm{PO}}\,d_t^2 + r_t, \qquad |r_t| \le \tfrac{L_3}{6}\,|d_t|^3,$$

with $L_3 = \sup|\Phi'''|$ on the operating interval. Summing over $t \le T$ and taking expectations, it remains to show $\mathbb{E}[\sum_t d_t] = 0$ under A2-sym. The recursion gives $\mathbb{E}[d_{t+1} \mid \mathcal{F}_t] = -m\,d_t + \mathbb{E}[u_t \mid \mathcal{F}_t] = -m\,d_t$, so $\mathbb{E}[d_{t+1}] = -m\,\mathbb{E}[d_t]$ by the tower property, and $\mathbb{E}[d_t] = (-m)^t\,\mathbb{E}[d_0] = 0$ when the loop starts at the operating point or in the stationary regime. The quadratic term contributes $\frac{1}{2}\gamma_{\mathrm{PO}}\,\mathbb{E}[\sum_t d_t^2]$, which is the claim.

For the variance decomposition, write the realized cost (net of the cubic remainder) as $C = -\delta_0 X + \frac{1}{2}\gamma_{\text{PO}} Y$ with $X = \sum_t d_t$, $Y = \sum_t d_t^2$. Then $\mathrm{Var}(C) = \delta_0^2 \,\mathrm{Var}(X) + \frac{\gamma_{\text{PO}}^2}{4}\,\mathrm{Var}(Y) - \delta_0 \gamma_{\text{PO}}\ \mathrm{Cov}(X,Y)$. Under Gaussian innovations the vector $(d_1, \dots, d_T)$ is jointly Gaussian with mean zero, and every third moment of a zero-mean Gaussian vector vanishes; since $\mathrm{Cov}(X,Y) = \sum_{t,s} \mathbb{E}[d_t\, d_s^2]$ is a sum of third moments, it is zero. (For non-Gaussian symmetric innovations the same conclusion follows from the sign symmetry $d \mapsto -d$ of the stationary law.) Verified: V0.1–V0.3, including the necessity check that an asymmetric rule breaks the expectation identity by exactly $-\delta_0\, \mathbb{E}[\sum_t d_t]$. $\square$

B. Saturation

Proof of Theorem T1 (information saturation). Let $\rho(M) < 1$ and define $P = \sum_{t \ge 0} (M^\top)^t M^t$. The series converges absolutely: by Gelfand's formula $\|M^t\| \le C\, \rho_0^t$ for any $\rho(M) < \rho_0 < 1$ and some C, so the terms are dominated by $C^2 \rho_0^{2t}$. Termwise, $M^\top P M = \sum_{t \ge 0} (M^\top)^{t+1} M^{t+1} = P - I$, so P solves the discrete Lyapunov equation, and if Q is any other solution then $P - Q = M^\top (P - Q) M = (M^\top)^k (P - Q) M^k \to 0$, giving uniqueness. The energy identity is immediate: $\sum_{t \ge 0} \|M^t d_0\|^2 = d_0^\top \big(\sum_t (M^\top)^t M^t\big) d_0 = d_0^\top P\, d_0$. The directional version replaces I by $uu^\top$: $P_u = \sum_t (M^\top)^t uu^\top M^t$ solves $P_u = uu^\top + M^\top P_u M$ and $\sum_t (u^\top M^t d_0)^2 = d_0^\top P_u\, d_0$.

For the information claim: under A3 the log-likelihood of the observations along the noiseless trajectory has Fisher information for the response component along u equal to $\sigma^{-2} \sum_t (u^\top d_t)^2 = \sigma^{-2}\, d_0^\top P_u d_0$, which is finite and independent of the horizon; the Cramér–Rao bound gives the stated variance floor for unbiased estimators (the biased/minimax version is Theorem T3's van Trees argument). Verified: V1.1, V1.2, V1.4. $\square$

Proof of Corollary C1.1 (safety implies blindness). In the scalar case $P = \sum_t m^{2t} = (1 - m^2)^{-1}$, so the cap $(h_0 - h^*)^2/(1 - m^2)$ has derivative in m equal to $2m\,(h_0 - h^*)^2/(1 - m^2)^2 > 0$ for $m \in (0, 1)$: strictly increasing. $\square$

Proof of Corollary C1.2 (design degeneracy). For $m < 1$, $d_t = (-m)^t d_0$, so for every $\epsilon > 0$ all but $\lceil \log(|d_0|/\epsilon)/\log(1/m) \rceil$ of the points lie in $[-\epsilon, \epsilon]$: the empirical design measure converges weakly to δ_0. At $m = 1$, $d_t = (-1)^t d_0$ and the design measure is $\frac{1}{2}(\delta_{+d_0} + \delta_{-d_0})$ exactly. In both cases the limiting design has at most two support points; combined with Theorem T6, the per-observation Fisher matrix of the three-parameter structural family is asymptotically singular along the trajectory. $\square$

Proof of Remark R1 (non-normal transient information). The identity $E(d_0) = d_0^\top P\, d_0$ of T1 holds for any stable M, normal or not. If $M = V \Lambda V^{-1}$ is diagonalizable, $\|M^t\| \le \kappa(V)\, \rho(M)^t$ gives $\|P\| \le \kappa(V)^2/(1 - \rho(M)^2)$. The bound is attained in order by the Jordan-type family $M = \begin{pmatrix} m & a \\ 0 & m \end{pmatrix}$ with $d_0 = e_2$: $M^t e_2 = (t\, a\, m^{t-1},\ m^t)^\top$, so

$$E(e_2) = a^2 \sum_{t \ge 1} t^2 m^{2t-2} + \sum_{t \ge 0} m^{2t} = a^2\, \frac{1 + m^2}{(1 - m^2)^3} + \frac{1}{1 - m^2},$$

using $\sum_{t \ge 1} t^2 x^{t-1} = (1 + x)/(1 - x)^3$ at $x = m^2$: the energy grows as a^2, without bound, at fixed spectral radius. Verified: V1.3 (the exact value matches the Lyapunov solve; 1268.2 vs the normal cap 2.78 at $m = 0.8$, $a = 6$). $\square$

C. The exchange rate

Proof of Theorem T2 (pathwise exchange rate). Under A3, conditionally on the design $\{d_t\}$ the observations are independent with means linear in ε and variance σ^2, so the least-squares slope has conditional variance $\mathrm{Var}(\hat{\varepsilon} \mid \text{design}) = \sigma^2/S_{xx}$ (Gauss–Markov). Multiplying by the quadratic cost $\frac{1}{2}\gamma_{\text{PO}} \sum_t d_t^2$ and using $\sum_t d_t^2 = S_{xx} + T\bar{d}^2$ gives the displayed identity exactly, on every realization. In the stationary regime $\bar{d} = O_P(T^{-1/2})$ by Proposition P2.1's long-run variance, while $S_{xx}/T \to v$ a.s., so $T\bar{d}^2/S_{xx} = O_P(1/T)$. Verified: V2.1 (machine precision). $\square$

Proof of Proposition P2.1 (concentration). For the stationary AR(1) with lag-one coefficient $\rho = -m$, $\operatorname{Cov}(d_t, d_s) = v\,(-m)^{|t-s|}$, so

$$\operatorname{Var}\Big(\sum_{t\le T} d_t\Big) = \sum_{t,s} v(-m)^{|t-s|} = Tv\,\frac{1+\rho}{1-\rho}\,(1+o(1)) = Tv\,\frac{1-m}{1+m}\,(1+o(1)),$$

the negative autocorrelation of the cobweb shrinking the fluctuation below the i.i.d. level. The realized-cost product then fluctuates around the constant with standard deviation $O(T^{-1/2})$ by Lemma L1's variance decomposition (both $\operatorname{Var}(\sum d)$ and $\operatorname{Var}(\sum d^2)$ are $O(T)$ while the squared mean of the cost is $O(T^2)$). Verified: V2.2 (variance ratio 4.80 for a $4\times$ horizon), V2.3. ☐

Proof of Proposition P2.2 (feedback bias). Under A3′, $d_{t+1} = -m\,d_t + \sigma_e\xi_t + \phi\,\zeta_t$, so d_t is a linear function of past innovations with $\mathbb{E}[d_s\zeta_t] = \phi\,\sigma^2\,(-m)^{s-t-1}$ for $s > t$ and 0 otherwise. Write the slope error as N/S with $N = \sum_t (d_t - \bar d)\,\zeta_t$ and $S = S_{xx}$, and expand $\mathbb{E}[N/S] = \mathbb{E}[N]/\mathbb{E}[S] - \operatorname{Cov}(N,S)/\mathbb{E}[S]^2 + O(T^{-2})$ (valid for stationary Gaussian data; the neglected term involves $\mathbb{E}[N]\operatorname{Var}(S)/\mathbb{E}[S]^3 = O(T^{-2})$).

Centering term. $\mathbb{E}[d_t\zeta_t] = 0$, so $\mathbb{E}[N] = -\frac{1}{T}\sum_{s>t}\mathbb{E}[d_s\zeta_t] = -\frac{1}{T}\sum_{s>t}\phi\sigma^2(-m)^{s-t-1} = -\phi\sigma^2/(1+m) + O(1/T)$.

Covariance term. For jointly Gaussian zero-mean (X,Y,Z), $\mathbb{E}[XY^2Z] = 2\,\mathbb{E}[XY]\mathbb{E}[YZ] + \mathbb{E}[XZ]\mathbb{E}[Y^2]$ (Isserlis). Applying this to $\operatorname{Cov}(N,S) = \sum_{t,s}\big(\mathbb{E}[\tilde d_t\tilde d_s^2\zeta_t] - \mathbb{E}[\tilde d_t\zeta_t]\mathbb{E}[\tilde d_s^2]\big)$ with $\tilde d = d - \bar d$ leaves only the paired term $2\sum_{t,s}\mathbb{E}[\tilde d_t\tilde d_s]\,\mathbb{E}[\tilde d_s\zeta_t]$; substituting the stationary covariances,

$$\operatorname{Cov}(N,S) = 2\,T\,v_\phi\,\phi\,\sigma^2\sum_{k\ge1}(-m)^k(-m)^{k-1} + O(1) = -\,\frac{2\,T\,v_\phi\,\phi\,\sigma^2\,m}{1-m^2} + O(1).$$

Combining with $\mathbb{E}[S] = Tv_\phi\,(1+o(1))$:

$$\mathbb{E}[\hat\varepsilon - \varepsilon_{\text{true}}] = \frac{\phi\sigma^2}{Tv_\phi}\Big[-\frac{1}{1+m} + \frac{2m}{1-m^2}\Big] + O(T^{-2}) = \frac{\phi\sigma^2\,(3m-1)}{(1-m^2)\,T\,v_\phi} + O(T^{-2}).$$

The bracket changes sign at $m = 1/3$. (The first draft kept only the centering term and was falsified by the verification suite; the record is in VERIFICATION.md.) Verified: V2.4, V2.5 – both signs, straddling $m = 1/3$. ☐

D. Minimax lower bounds

Proof of Theorem T3 (deviation-budget minimax). Let π be a prior density on an interval of width w with finite information $I(\pi) = \int(\pi')^2/\pi$; the cosine-squared density attains the minimum $I(\pi) = (\pi/w)^2$ (here the first π is the constant). For any non-anticipating policy the joint density of the data factorizes as $\prod_t p(u_t \mid \mathcal{F}_t)\,p(\tau_t \mid \mathcal{F}_t, \varepsilon)$, where the policy factors do not depend on ε and $\tau_t \mid \mathcal{F}_t \sim \mathcal{N}(\tau(h^*) - \varepsilon\,d_t, \sigma^2)$ with d_t being $\mathcal{F}_t$-measurable. Hence the Fisher information of the whole experiment is

$$I_T(\varepsilon) = \mathbb{E}_\varepsilon\Big[\sum_t \big(\partial_\varepsilon \log p(\tau_t \mid \mathcal{F}_t, \varepsilon)\big)^2\Big] = \mathbb{E}_\varepsilon\Big[\sum_t d_t^2\Big]/\sigma^2 \;\le\; D/\sigma^2$$

under the pathwise budget: adaptivity places information but cannot manufacture it. The van Trees inequality (Gill–Levit) then bounds the Bayes risk of *any* estimator, biased or not: $\mathbb{E}_\pi\mathbb{E}[(\hat\varepsilon - \varepsilon)^2] \ge \big(\mathbb{E}_\pi[I_T] + I(\pi)\big)^{-1} \ge \big(D/\sigma^2 + (\pi/w)^2\big)^{-1}$, and the supremum over the interval dominates the prior average. Equality (up to the prior term) is attained by symmetric two-point probing at the budget with least squares, by Theorem T2. Verified: V3.1, V3.2 (equality within Monte-Carlo error). ☐

Proof of Lemma L2 (exploitation–information). Prior $s = \pm1$ symmetric; environments $P^\pm$ differing only in the observation means by $2\delta\,d_t$ at step t. Three ingredients.

(i) *Posterior imbalance equals total variation in expectation.* Under the symmetric mixture $\bar P = \frac{1}{2}(P^+ + P^-)$, with data x,

$$\mathbb{E}_{\bar P}\big|\mathbb{P}(s{=}+ \mid x) - \mathbb{P}(s{=}- \mid x)\big| = \int \tfrac{1}{2}(dP^+ + dP^-)\,\frac{|dP^+ - dP^-|}{dP^+ + dP^-} = \operatorname{TV}(P_t^+, P_t^-).$$

(ii) *Pinsker with the chain rule for adaptive designs.* $\mathrm{KL}(P_t^+ \,\|\, P_t^-) = \mathbb{E}^+\big[\sum_{i\le t}(2\delta\, d_i)^2/(2\sigma^2)\big] = 2\delta^2\,\mathbb{E}^+[S_t]/\sigma^2$ (the policy factors cancel inside the log-ratio), so $\mathrm{TV}_t \le \delta\,\sqrt{\mathbb{E}[S_t]}/\sigma$.

(iii) *The gain bound.* Since d_t is $\mathcal{F}_t$-measurable, $\mathbb{E}[s\,d_t] = \mathbb{E}[d_t\,\mathbb{E}[s \mid \mathcal{F}_t]]$; two applications of Cauchy–Schwarz and $|\mathbb{E}[s \mid \mathcal{F}_t]| \le 1$ give

$$\mathbb{E}[\mathrm{gain}_T] = g_1\delta\sum_t \mathbb{E}[s\,d_t] \le g_1\delta\sum_t \big(\mathbb{E}d_t^2\big)^{1/2}\,\mathrm{TV}_t^{1/2} \le g_1\delta\,\big(\mathbb{E}S_T\big)^{1/2}\,\big(T\,\mathrm{TV}_T\big)^{1/2},$$

and inserting (ii): $\mathbb{E}[\mathrm{gain}_T] \le g_1\,\delta^{3/2}\,T^{1/2}\,(\mathbb{E}S_T)^{3/4}/\sigma^{1/2}$.

Remark on constants. The derivation document's heuristic chain (treating the imbalance as bounded by TV_t pathwise) yields $g_1\delta^2 S_T\sqrt{T}/\sigma$; the rigorous chain above changes the exponent bookkeeping but not the conclusion of Theorem T4: at the minimax scale $\delta = c\,\sigma/\sqrt{\mathbb{E}S_T}$ both give $\mathbb{E}[\mathrm{gain}_T] = O(\sigma\sqrt{T}\,)$, hence an exploited fraction $O(T^{-1/2})$ of the $\Theta(T)$ cost under A5. Verified: V3.3 (the imbalance–TV inequality pathwise in expectation), V3.4 (the $T^{-1/2}$ decay at the minimax scale). ☐

Proof of Theorem T4 (value-budget minimax; constant-factor version). Fix a policy and estimator; let $\bar S = \mathbb{E}[S_T] = \Theta(T)$ under A5. Choose the two-point prior at separation $\delta = c\,\sigma/\sqrt{\bar S}$ with $c \in (0,1)$ to be optimized. By the testing reduction (Le Cam), any estimator's worst-case risk over the pair obeys $\sup_\pm \mathbb{E}[(\hat\varepsilon - \varepsilon)^2] \ge \frac{\delta^2}{2}(1 - \mathrm{TV}_T) \ge \frac{\delta^2}{2}(1-c)$ using Lemma L2(ii). By Lemma L1 with the certainty-equivalent anchor and Lemma L2's gain bound, the value cost obeys $C_T \ge \frac{1}{2}\gamma_{\mathrm{PO}}\,\bar S\,(1 - O(T^{-1/2}))$. Multiplying,

$$\sup\ \mathrm{Var}(\hat\varepsilon)\times C_T \;\ge\; \frac{c^2\sigma^2}{2\bar S}(1-c)\cdot\frac{\gamma_{\mathrm{PO}}\bar S}{2}\,\big(1-O(T^{-1/2})\big) \;=\; \frac{\gamma_{\mathrm{PO}}\sigma^2}{4}\,c^2(1-c)\,\big(1-O(T^{-1/2})\big),$$

maximized at $c = 2/3$: the bound $\gamma_{\mathrm{PO}}\sigma^2/27\cdot(1 - O(T^{-1/2}))$.

Status. This proves the value-budget product is bounded below by an explicit constant multiple of the exchange rate, uniformly over adaptive policies, with the exploited rebate vanishing at rate $T^{-1/2}$. The sharp constant $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$ – numerically supported (V3.4: no simulated policy fell below it) – requires the van Trees route with cost control in place of the two-point reduction and is part of OPEN-1. ☐

E. Design geometry

Proof of Theorem T5 (optimal exploration shapes). All three problems have convex (resp. concave) objectives over the PSD cone with one linear constraint; KKT stationarity plus convexity/concavity certifies global optimality.

(a) *A-optimality.* $\nabla_M \operatorname{tr}(M^{-1}) = -M^{-2}$, so stationarity gives $M^{-2} = \lambda\,\Gamma_{\mathrm{PO}}$, i.e. $M = \lambda^{-1/2}\Gamma_{\mathrm{PO}}^{-1/2}$; the budget $\operatorname{tr}(\Gamma_{\mathrm{PO}}M) = \lambda^{-1/2}\operatorname{tr}(\Gamma_{\mathrm{PO}}^{1/2}) = B$ pins $\lambda^{-1/2} = B/\operatorname{tr}(\Gamma_{\mathrm{PO}}^{1/2})$ and the value $\operatorname{tr}(M^{-1}) = (\operatorname{tr}\Gamma_{\mathrm{PO}}^{1/2})^2/B$. (Diagonal-case cross-check by Cauchy–Schwarz: $(\sum_i 1/m_i)(\sum_i g_i m_i) \ge (\sum_i \sqrt{g_i})^2$ with equality at $m_i \propto g_i^{-1/2}$.)

(b) *D-optimality.* $\nabla_M \log\det M = M^{-1} = \lambda\,\Gamma_{\mathrm{PO}}$ gives $M = (B/d)\,\Gamma_{\mathrm{PO}}^{-1}$ after normalizing by the budget.

(c) *c-optimality.* Substitute $N = \Gamma_{\mathrm{PO}}^{1/2} M\,\Gamma_{\mathrm{PO}}^{1/2}$ ($\operatorname{tr} N = \operatorname{tr}(\Gamma_{\mathrm{PO}}M) \le B$) and $w = \Gamma_{\mathrm{PO}}^{1/2}c$, so the objective is $w^\top N^{-1} w$. By Cauchy–Schwarz, $(w^\top w)^2 = (w^\top N^{-1/2}\,N^{1/2}w)^2 \le (w^\top N^{-1}w)(w^\top N w)$ and $w^\top N w \le \|w\|^2 \operatorname{tr} N \le \|w\|^2 B$, hence $w^\top N^{-1} w \ge \|w\|^2/B = c^\top\Gamma_{\mathrm{PO}}\,c/B$, attained at $N^* = B\,ww^\top/\|w\|^2$. Back-substituting with $\Gamma_{\mathrm{PO}}^{-1/2}w = c$ gives $M^* = B\,cc^\top/(c^\top\Gamma_{\mathrm{PO}}\,c)$: the optimal probe direction is c itself, the curvature pricing only its scale. Verified: V4.1–V4.3 (formulas achieved; never beaten by 4000 random feasible designs each). ☐

Proof of Corollary C5.1 (price of isotropic jitter). Isotropic $M = (B/\operatorname{tr}\Gamma_{\mathrm{PO}})\,I$ has $\operatorname{tr}(M^{-1}) = d\,\operatorname{tr}(\Gamma_{\mathrm{PO}})/B$; dividing by the optimum $(\operatorname{tr}\Gamma_{\mathrm{PO}}^{1/2})^2/B$ gives $F = d\,\operatorname{tr}(\Gamma_{\mathrm{PO}})/(\operatorname{tr}\Gamma_{\mathrm{PO}}^{1/2})^2$, and $F \ge 1$

with equality iff all eigenvalues are equal, by Cauchy–Schwarz applied to $(\sum_i \sqrt{g_i} \cdot 1)^2 \le d \sum_i g_i$. Verified: V4.4. $\square$

Proof of Lemma L3 (temporal shaping). Under A3 the conditional log-likelihood of the responses given the deployment path is $-\sum_t (\tau_t - a - \theta^\top d_t)^2/(2\sigma^2)$ up to constants; its Hessian in θ is $-\sum_t d_t d_t^\top/\sigma^2$, a function of the empirical second moment only, whatever the temporal dependence of $\{d_t\}$. Verified: V4.5. $\square$

Proof of Theorem T6 (Chebyshev unidentifiability). The sensitivity of the family $\tau(h) = C_0 + C_1 e^{-ch}$ is $s(h) = (1,\ e^{-ch},\ -C_1 h e^{-ch})^\top$. The functions $\{1, e^{-ch}, he^{-ch}\}$ form an extended Chebyshev system on any interval (their Wronskian is $c^3 e^{-2ch} \neq 0$ after row reduction; equivalently, a nontrivial linear combination $a + (b + \gamma h)e^{-ch}$ has at most two zeros counting multiplicity). For any design ξ with k support points, the Fisher matrix $M(\xi) = \sum_{i \le k} w_i\, s(h_i)s(h_i)^\top$ has rank at most k; with $k \le 2 < 3$ it is singular, and its null space necessarily has a component along the c-sensitivity (otherwise the two-point system would determine all three parameters, contradicting the Chebyshev property). By Corollary C1.2 the trajectory design has at most two effective support points, which proves the claim; the existence of exactly-three-point D-optimal designs on a trust-region interval is classical for Chebyshev systems (Karlin–Studden). Verified: V5.1 (two-point determinants at 10^{-17}; three-point at 6×10^{-3}). $\square$

F. Safety under certainty equivalence

Proof of Lemma L4 (perturbed modulus). The corrected map is $\Psi(h) = h + \eta\,[\,G(h) - \beta\,(h - \psi)\,\hat{\varepsilon}(h)\,]$ and the ideal map replaces $\hat{\varepsilon}$ by ε. With $e = \hat{\varepsilon} - \varepsilon$, $\Psi'(h) - \Psi'_{\text{ideal}}(h) = -\eta\,\beta\,\frac{d}{dh}\big[(h - \psi)\,e(h)\big] = -\eta\,\beta\,\big[e(h) + (h - \psi)e'(h)\big]$, and taking suprema over the operating interval gives $|\hat{\rho} - \rho| \le \eta\beta\,(\|e\|_\infty + |h - \psi|_{\max}\|e'\|_\infty)$. Verified: V6.1 (nine-cell error grid; the bound holds with maximal excess 2.4×10^{-11}). $\square$

Proof of Remark R2 (open-loop neutrality). For $d_{t+1} = -m\,d_t + u_t$ with a history-independent schedule $\{u_t\}$, two solutions with different initial conditions differ by the homogeneous solution $(-m)^t(d_0 - d_0')$, which contracts at rate m regardless of the schedule (superposition). Verified: V6.2 (difference exactly 0 after 200 steps to machine precision). $\square$

Proof of Proposition P7.1 (safety under pessimism). Let E be the event that the confidence sequence contains the true response parameters simultaneously for all t; by anytime-validity $\mathbb{P}(E) \ge 1 - \delta$. On E, the true parameters lie in Θ_t at every step, so the realized modulus is dominated by the rule's worst case: $|\rho_t| \le \sup_{\theta \in \Theta_t} |\hat{\rho}(\theta)| \le 1 - c_{\text{margin}}$ whenever correction is active, and when the rule freezes, the blind loop's modulus $m < 1 - c_{\text{margin}}$ (choose the margin below $1 - m$) applies. No union bound over time is needed: the inclusion is pathwise on E. *(Conditional result: the hypothesis is the confidence sequence's validity; constructing it from the D2 information accounting is routine, and the necessity of the hypothesis is demonstrated numerically – an invalid, too-narrow sequence produces violations.)* $\square$

G. The anchoring crossover

Proof of Theorem T7 (horizon-dependent crossover). Bias. For the symmetric secant with half-width w, fifth-order Taylor expansion of $\tau(h^* \pm w)$ gives

$$\frac{\tau(h^* + w) - \tau(h^* - w)}{2w} = \tau'(h^*) + \frac{\tau'''(h^*)}{6}\, w^2 + O(w^4),$$

all even-order terms cancelling (symmetric probing is bias-optimal at its order; the second derivative never enters). Verified: V7.1.

Variance. With n probe pairs ($T = 2n$ observations) the slope estimator $(\bar{y}_+ - \bar{y}_-)/(2w)$ has variance $(2\sigma^2/n)/(4w^2) = \sigma^2/(n\,w^2\,2) \cdot \ldots = \sigma^2/S$ with $S = \sum_t d_t^2 = 2nw^2 = Tw^2$: variance depends on the design only through its energy.

Constrained optimum. The value budget fixes $S = 2B/\gamma_{\mathrm{PO}}$; the horizon caps the number of observations, $Tw^2 \geq S$, i.e. $w^2 \geq 2B/(\gamma_{\mathrm{PO}}T)$; bias is increasing in w, so the constrained optimum sits at $w^2 = 2B/(\gamma_{\mathrm{PO}}T)$, giving

$$\mathrm{MSE}_{\mathrm{np}} = \frac{\gamma_{\mathrm{PO}}\,\sigma^2}{2B} + \Big(\frac{\tau'''(h^*)}{6}\Big)^2\Big(\frac{2B}{\gamma_{\mathrm{PO}}T}\Big)^2.$$

Verified: V7.2–V7.4 (three (B,T) cells, within 2%).

Crossover. The anchored estimator has $\mathrm{MSE} = \kappa_p\,\gamma_{\mathrm{PO}}\sigma^2/(2B) + \delta_{\mathrm{mis}}^2$; comparing and keeping the dominant term gives the rule $\delta_{\mathrm{mis}} < |\tau'''(h^*)|\,B/(3\gamma_{\mathrm{PO}}T)$ to leading order, with the exact two-term form in the display above. The direction of the comparison on both sides of the threshold is verified: V7.5, V7.6. ☐

H. ROI and instantiation

Proof of Theorem T8 (ROI of self-knowledge). With $A = \frac{1}{2}\gamma_{\mathrm{PO}}(h_{\mathrm{SP}} - h_{\mathrm{PO}})^2$, $a = \frac{1}{2}\gamma_{\mathrm{PO}}\kappa^2$, $b = \frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$, the net present value of exploring to accuracy v is $\mathrm{NPV}(v) = (A - av)/\rho_{\mathrm{disc}} - b/v$, strictly concave on $(0,\infty)$ with $\mathrm{NPV}'(v) = -a/\rho_{\mathrm{disc}} + b/v^2$, giving $v^* = \sqrt{b\,\rho_{\mathrm{disc}}/a} = (\sigma/\kappa)\sqrt{\rho_{\mathrm{disc}}}$ (the curvature cancels between a and b). Substituting, $\mathrm{NPV}(v^*) = A/\rho_{\mathrm{disc}} - 2\sqrt{ab/\rho_{\mathrm{disc}}}$, which is positive iff $A > 2\sqrt{ab\,\rho_{\mathrm{disc}}}$, i.e. iff $\rho_{\mathrm{disc}} < A^2/(4ab) = (h_{\mathrm{SP}} - h_{\mathrm{PO}})^4/(4\kappa^2\sigma^2)$: γ_{PO} cancels completely. Verified: V8.1, V8.2 (the break-even root to 10^{-3} relative). ☐

Proof of Proposition P9.1 (frontier consistency). For any deterministic design, $\mathrm{Var}(\hat{\varepsilon} \mid \mathrm{design}) \cdot \frac{1}{2}\gamma_{\mathrm{PO}}S = (\sigma^2/S)\cdot\frac{1}{2}\gamma_{\mathrm{PO}}S = \frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$: the design energy cancels identically, whatever the schedule – including the $d_t \propto t^{-1/2}$ schedules with $S_T = \Theta(\log T)$ of the Lai–Robbins adaptive-design line. Verified: V8.3 (exact). ☐

Proof of Proposition P9.2 (isotropy contrast). Immediate from Corollary C5.1: $F = 1$ iff the curvature spectrum is flat – the isotropic (LQR-like) case in which naive exploration is optimal; any curvature dispersion makes $F > 1$ by the strict case of Cauchy–Schwarz. Cross-verified with V4.4. ☐

Proof of Proposition P9.3 (separable multi-bond curvature). In the separable case the objective is a sum of per-bond objectives $\Phi_a(h_a)$, each of the scalar form analyzed in the base project's theory 1.2, so the Hessian is diagonal and the scalar computation applies per bond: differentiating $\Phi_a(h) = P[\,hA_a e^{-k_a h}\rho + h\tau_a(h) - \psi_a\tau_a(h) - w_a(h - h_{\mathrm{ref}})^2\,]$ twice and collecting terms with $\tau_a' = -\varepsilon_a$, $\tau_a'' = c_t\,\varepsilon_a$ yields $-\Phi_a'' = \gamma_a + \beta\,\varepsilon_a\,(2 + c_t\psi_a - c_t h_a)$, which is the claimed diagonal entry. In the factor-coupled case the off-diagonal response contributes additional terms of first order in $\|E_{\mathrm{offdiag}}\|$, stated as a construction with its error term (not claimed exact). Verified: V8.4 (finite-difference Hessians of three random bonds, relative error below 10^{-4}). ☐

C Selected verification rows

D Additional figures

E Reproducibility

All results are deterministic from seeds, numpy-only, CPU-only. The repository contains: the derivation documents and theorem register (25 results; 34 numerical derivation checks), the pipeline (environments, estimators, designs, agents, baselines; 36 measured-vs-predicted rows; 9 unit tests), the real-data leg with its 10/10 port validation, and continuous integration running the full verification on every push. The register of ten measurement-forced pivots — each recorded in the code where it happened, with the failed prediction and the fix — is in the repository README.

Table 1: Selected verification rows (full table, seeds, and code in the repository).

Claim (register)	Measured	Predicted	
Exchange rate, m=0.3, two amplitudes (T2)	$0.01062,\ 0.01070$	0.01056	pass
Exchange rate, m=0.6, two amplitudes (T2)	$0.00549,\ 0.00569$	0.00535	pass
Saturating environment (outside A1)	$+16\%$ over rate	—	drift
Feedback floor, m=0.3/0.6/0.85 (T1′)	within 1–7%	$(\sigma/\gamma)^2/(1-m^2)$	pass
Design arm $\mathrm{sd}(\hat{c})$; jitter/design ratio (T6)	0.234; 7.8×	Fisher 0.248; ≥ 3	pass
SafeD reaches h_{PO}; trust region (L4)	1.324; 0 violations	1.296; 0	pass
Isotropic/A-optimal risk ratio, d=8 (T5)	1.529	$F = 1.506$	pass
Nonparametric MSE at (B, T) optimum (T7)	0.0132	0.0123	pass
Pricing domain, 4 cells (T2, exact: LQ)	$0.384,\ 0.192$	$0.384,\ 0.192$	pass
Multi-bond shaped/iso steady-state risk (T5)	1.20 (flat control 1.00)	$> 1.15\ (\approx 1)$	pass
Structure-proof floor, min CR ratio (OPEN-1)	1.005	≥ 1	pass

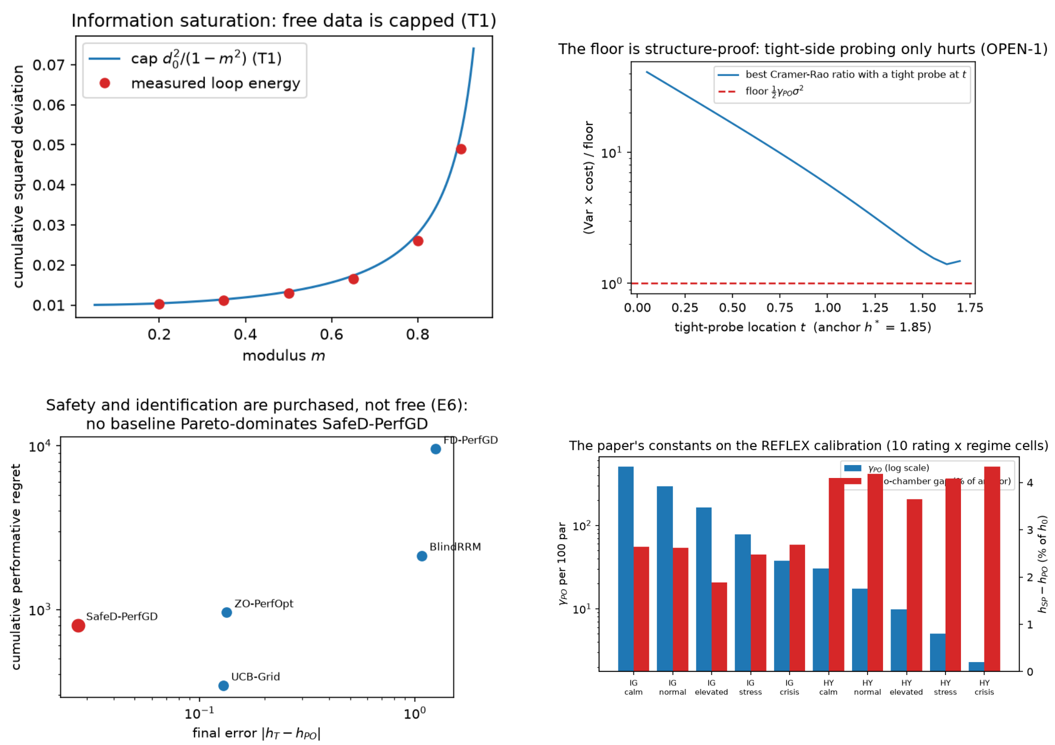

Figure 3: **Top left:** information saturation — measured loop energy against the cap $d_0^2/(1-m^2)$ (T1). **Top right:** the floor is structure-proof — best Cramér–Rao ratio with a tight probe at t; tight-side probing only hurts (OPEN-1). **Bottom left:** baseline Pareto frontier — no baseline dominates SafeD-PerfGD on (final error, regret). **Bottom right:** the paper's constants on the real-data calibration, 10 rating×regime cells.